

ON THE BOUNDARY CONDITION ON THE EARTH SURFACE FOR DIFFUSING POLLUTION

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The diffusion of pollution along the vertical will be described here by means of the routine parabolic equation

$$(1) \quad \frac{\partial q}{\partial t} - w \frac{\partial q}{\partial z} = \frac{\partial}{\partial z} K(z) \frac{\partial q}{\partial z}$$

where q is the concentration of the pollution, t the time, w the velocity of gravitational fall of pollution particles, and $K(z)$ the coefficient of vertical diffusion. The vertical pollution flux is then $K(z) \frac{\partial q}{\partial z} + wq$. The boundary condition on the earth surface must reflect the character of the interaction between the diffusing particles and the surface; such a condition can be written as

$$(2) \quad \left(K \frac{\partial q}{\partial z} + wq - \beta q \right)_{z=z_0} = 0$$

where β is a constant characterizing the interaction between the diffusing pollution and the surface. When $\beta = 0$ the condition (2) means that the pollution flux through the earth surface is zero, so that the entire pollution remains in the air; in other words, the pollution particles are reflected from the earth surface. When $\beta = \infty$ the condition (2) takes the form $(q)_{z=z_0} = 0$; it means that the pollution particles reaching the earth surface stick to the earth or are absorbed by it.

When $0 < \beta < \infty$ the condition (2) means that the particles reaching the earth surface can be either reflected from or absorbed by it. In order to find the statistical meaning of β , let us consider the simplest case $K(z) = \text{const}$, $z_0 = 0$. The solution of Equation (1) when there is an instantaneous point source of unit intensity at the height $z = h$ and at the moment $t = 0$ is, under the boundary condition (2),

$$(3) \quad q(z, t; h) = \frac{1}{2\sqrt{(\pi K t)}} e^{-w(z-h)/(2K) - w^2 t/(4K)} [e^{-(z-h)^2/(4Kt)} + e^{-(z+h)^2/(4Kt)}] - \frac{K}{2\beta - w} e^{\beta h/K + (\beta - w)z/K + \beta(\beta - w)t/K} \phi \left[-\frac{z + h + (2\beta - w)t}{\sqrt{(2Kt)}} \right]$$

where $\phi(x)$ is the standardized normal integral probability distribution. The fraction of the diffusing particles to be absorbed by the earth surface at the moment t is determined by the expression

$$(4) \quad P(t; h) = 1 - \int_0^\infty q(z, t; h) dz = \frac{\beta}{\beta - w} e^{wh/K} \phi\left(-\frac{h + wt}{\sqrt{(2Kt)}}\right) \\ + \phi\left(-\frac{h - wt}{\sqrt{(2Kt)}}\right) - \frac{2\beta - w}{\beta - w} e^{\beta h/K + \beta(\beta - w)t/K} \\ \times \phi\left[-\frac{h + (2\beta - w)t}{\sqrt{(2Kt)}}\right].$$

The value $P(t; h)$ is the probability of the event that the particle, being at the moment $t = 0$ in the point $z = h$, will be absorbed during the time t by the earth surface. In particular, the particle reaching the earth surface at the moment $t = 0$, will be absorbed by it during the time t with the probability

$$(5) \quad P(t; 0) = 1 + \frac{w}{\beta - w} \phi\left(-w\sqrt{\frac{t}{2K}}\right) - \frac{2\beta - w}{\beta - w} e^{\beta(\beta - w)t/K} \phi\left[(w - 2\beta)\sqrt{\frac{t}{2K}}\right]$$

When t is small this expression takes the form

$$(6) \quad P(t; 0) = 2\beta\sqrt{\left(\frac{t}{\pi K}\right)} + O(\sqrt{t})$$

so that the value β can be determined by

$$(7) \quad \beta = \lim_{t \rightarrow 0} \frac{\sqrt{(\pi K)} P(t; 0)}{2\sqrt{t}}$$

This formula throws light on the statistical meaning of the parameter β . Note that if the process of the absorption of the particles obeyed the Poisson law, then $P(t, 0)$ if t is small would be proportional to t and not to $\sqrt{t}$. Consequently, the boundary condition (2) implies that the absorption of particles reaching the earth surface occurs more often than in the case of the Poisson law.